

Inborn Errors of Metabolism

Jason Ryan, MD, MPH

Inborn Errors in Metabolism

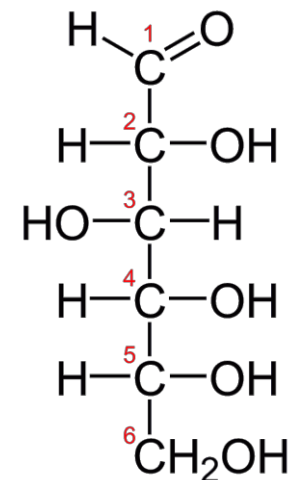
- Defects in metabolic pathways
- Often present in newborn period
- Often non-specific features:
 - Failure to thrive, hypotonia
- Lab findings suggest diagnosis:
 - Hypoglycemia
 - Ketosis
 - Hyperammonemia
 - Lactic acidosis



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Newborn Hypoglycemia

- Glycogen storage diseases
- Galactosemia
- Hereditary fructose intolerance
- Organic acidemias
- Disorders of fatty acid metabolism



Glucose

Glycogen Storage Diseases

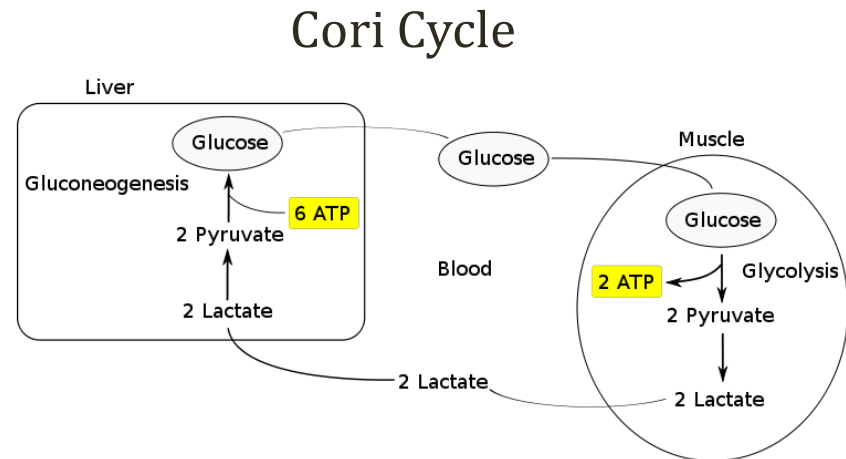
- Some have **no hypoglycemia**
 - Only affect muscles
 - McArdle's Disease (type V)
 - Pompe's Disease (type II)
- Hypoglycemia seen in others
 - Von Gierke's Disease (Type I)
 - Cori's Disease (Type III)

Glycogen Storage Diseases

- Fasting hypoglycemia
 - Hours after eating
 - Not in post-prandial period
- **Ketosis**
 - Absence of glucose during fasting
 - Fatty acid breakdown (NOT a fatty acid disorder)
 - Ketone synthesis
- Hepatomegaly
 - Glycogen buildup in liver

Glycogen Storage Diseases

- Von Gierke's Disease (Type I)
 - Severe hypoglycemia
 - Lactic acidosis
- Cori's Disease (Type III)
 - Gluconeogenesis intact
 - Mild hypoglycemia
 - No lactic acidosis

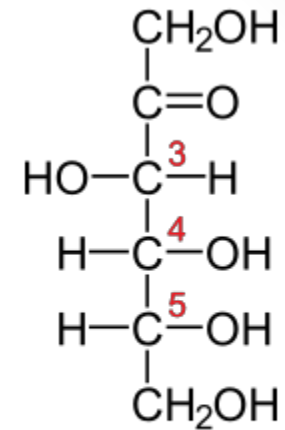


Petaholmes/Wikipedia

HFI

Hereditary Fructose Intolerance

- Deficiency of **aldolase B**
- Build-up of fructose 1-phosphate
- Depletion of ATP
- Loss of gluconeogenesis and glycogenolysis
- Hypoglycemia
- Lactic acidosis
- Ketosis
- Hepatomegaly (glycogen buildup)



D-Fructose

HFI

Hereditary Fructose Intolerance

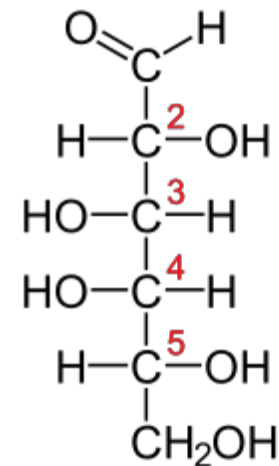
- Starts after **weaned from breast milk**
 - No fructose in breast milk
- **“Reducing sugars”** in urine
 - Glucose, fructose, galactose
 - Reducing sugars in urine with hypoglycemia



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Classic Galactosemia

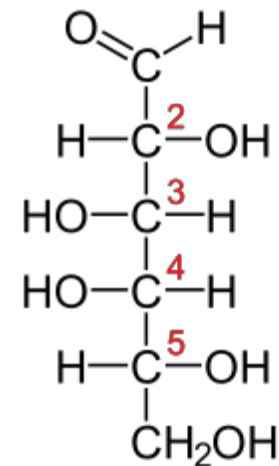
- Deficiency of galactose 1-phosphate uridylyltransferase
- Galactose-1-phosphate accumulates
- Depletion of ATP
- 1st few days of life
- Breast milk contains lactose
- Lactose = galactose + glucose



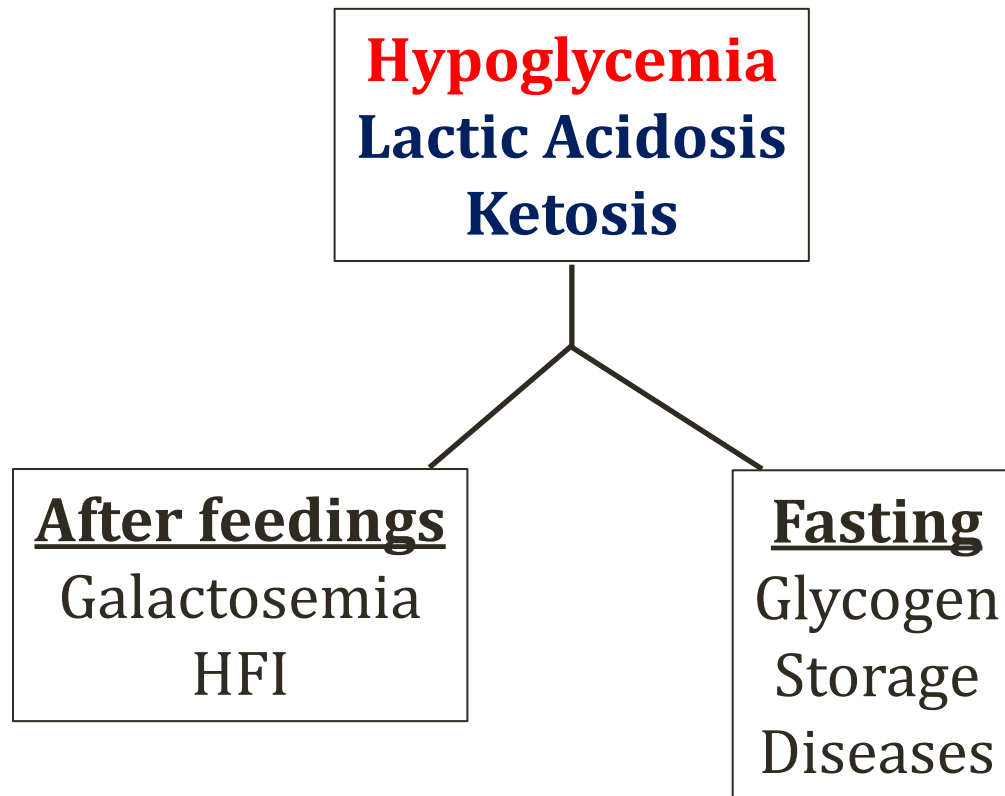
D-Galactose

Classic Galactosemia

- **Vomiting/diarrhea after feeding**
- Similar presentation to HFI
 - Hypoglycemia
 - Lactic acidosis
 - Ketosis
 - Hepatomegaly (glycogen buildup)
 - “Reducing sugars” in urine

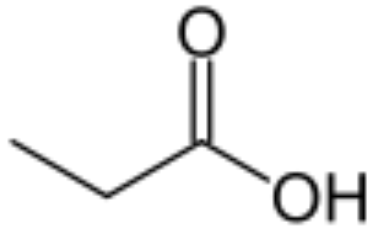


D-Galactose

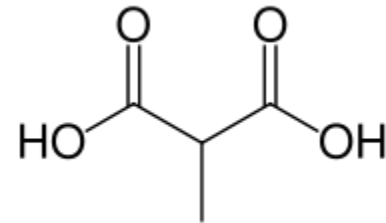


Organic Acidemias

- Abnormal metabolism of organic acids
 - Propionic acid
 - Methylmalonic acid
- Buildup of organic acids in blood/urine
- **Hyperammonemia**



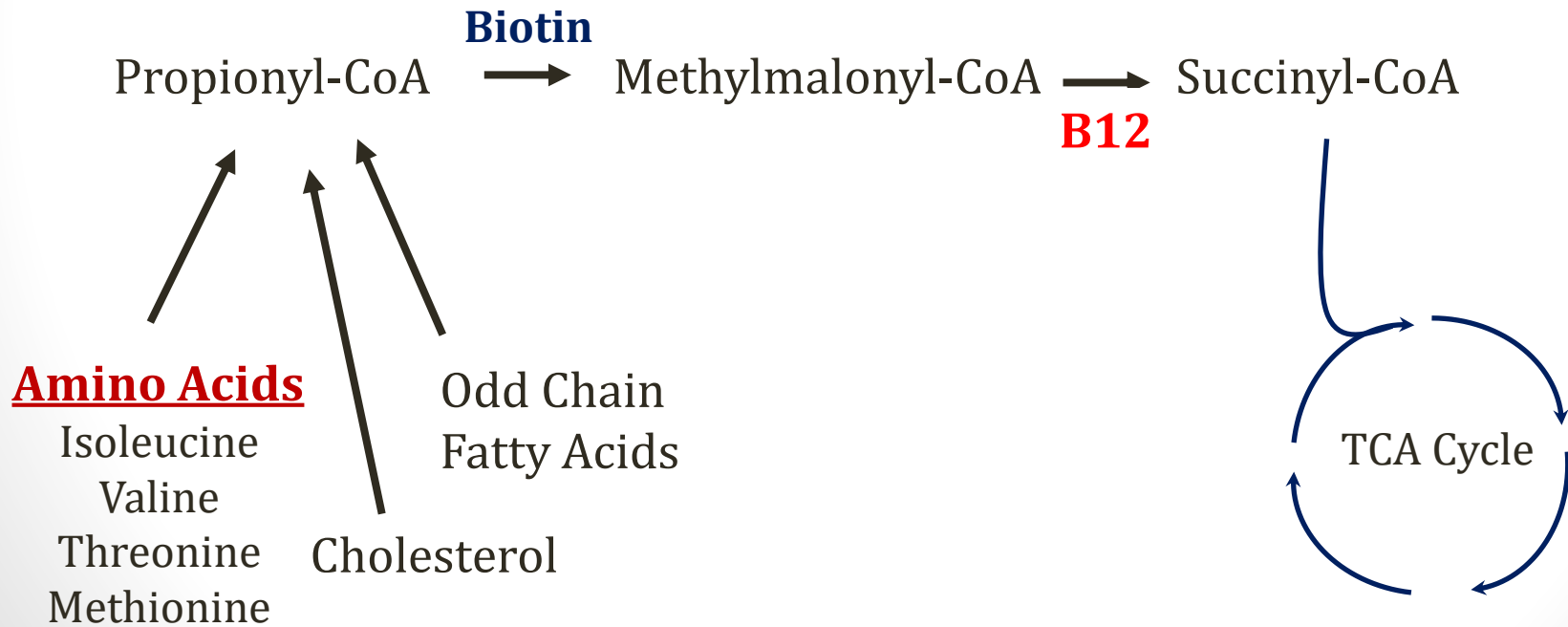
Propionic Acid



Methylmalonic Acid

Succinyl-CoA

- Common pathway to TCA cycle
- Many substances metabolized to propionyl-CoA
- Propionyl-CoA $\rightarrow$ Methylmalonyl-CoA

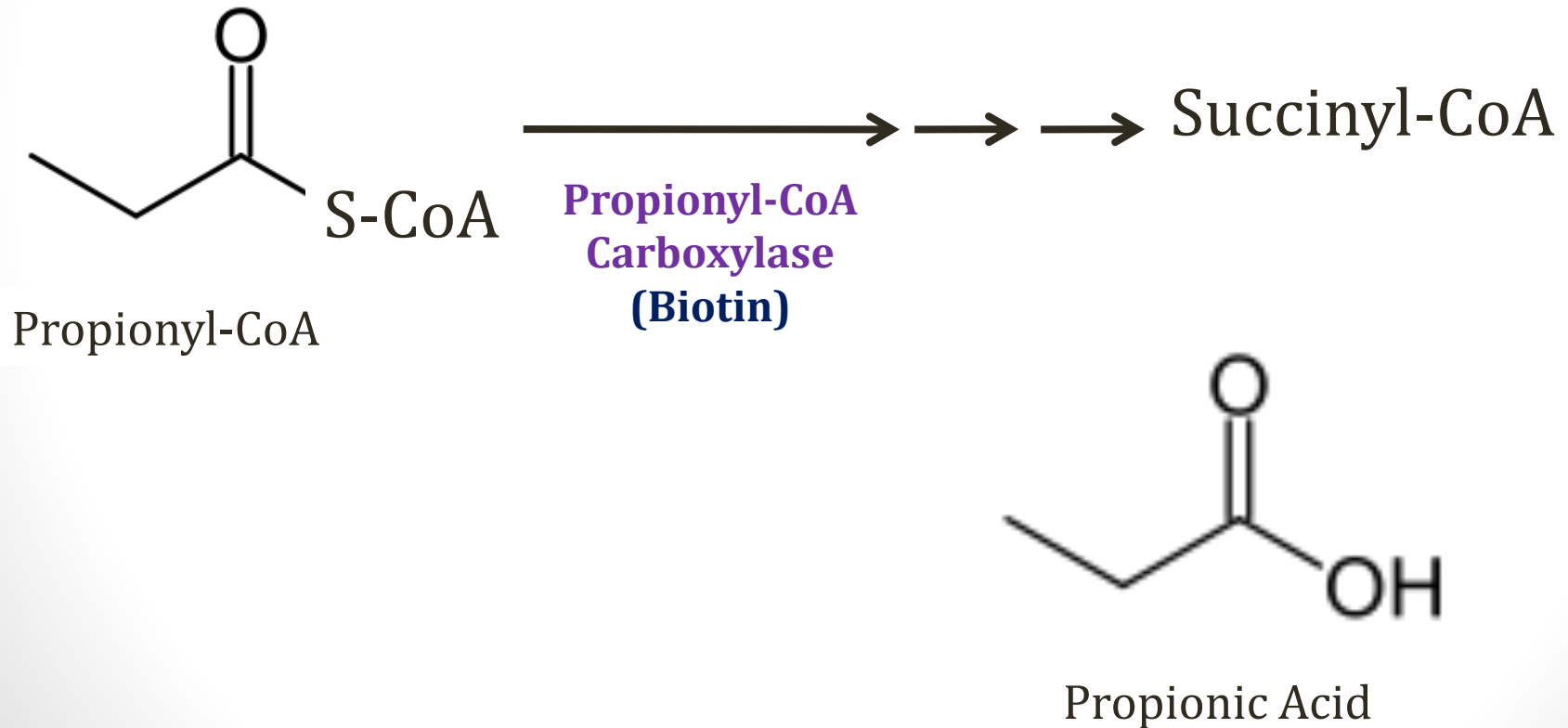


Organic Acidemias

- Onset in newborn period (weeks-months)
- Poor feeding, vomiting, hypotonia, lethargy
- Hypoglycemia → ketosis
 - Complex mechanism
 - Liver damage → ↓ gluconeogenesis
- Anion gap metabolic acidosis
- **Hyperammonemia**
- **Elevated urine/plasma organic acids**

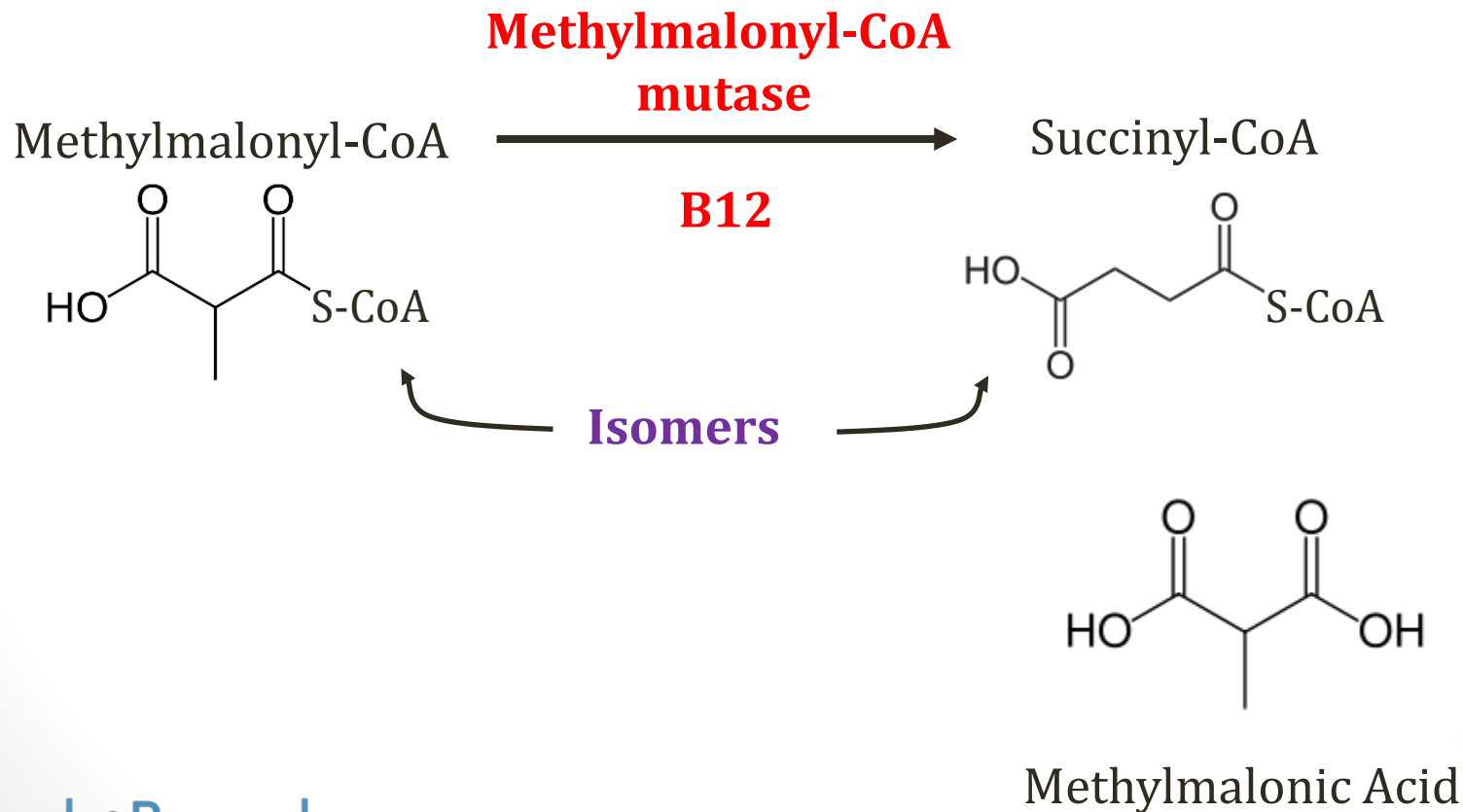
Propionic Acidemia

- Deficiency of **propionyl-CoA carboxylase**



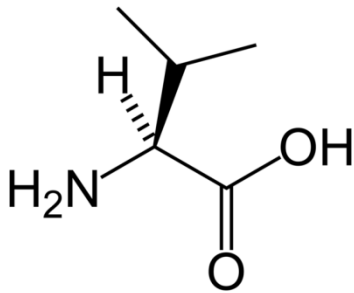
Methylmalonic Acidemia

- Deficiency of **methylmalonyl-CoA mutase**

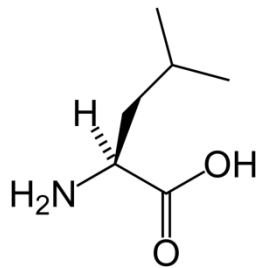


Maple Syrup Urine Disease

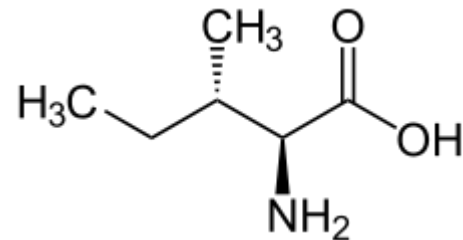
- Branched chain amino acid disorder
- Deficiency of **α -ketoacid dehydrogenase**
 - Multi-subunit complex
 - Cofactors: Thiamine, lipoic acid
- Amino acids and α -ketoacids in plasma/urine
- α -ketoacid of isoleucine gives urine sweet smell



Valine



Leucine



Isoleucine

Fatty Acid Disorders

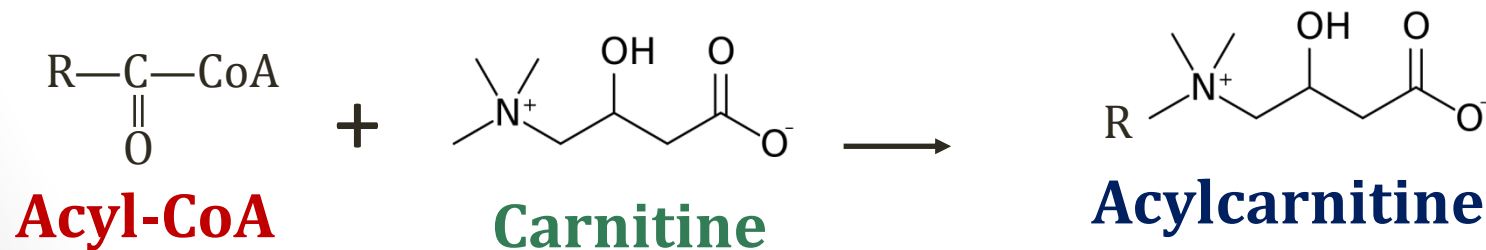
- Carnitine deficiency
- MCAD deficiency
 - Medium-chain-acyl-CoA dehydrogenase
 - Beta oxidation enzyme
- Both cause **hypoketotic hypoglycemia** when fasting
 - Lack of fatty acid breakdown → low ketone bodies
 - Overutilization of glucose → hypoglycemia
 - Lack of acetyl-CoA for gluconeogenesis

Fatty Acid Disorders

- Symptoms with fasting or illness
- Usually 3 months to 2 years
 - Frequent feedings < 3 months prevent fasting
- Failure to thrive, altered consciousness, hypotonia
- Hepatomegaly
- Cardiomegaly
- **Hypoketotic hypoglycemia**

Primary Carnitine Deficiency

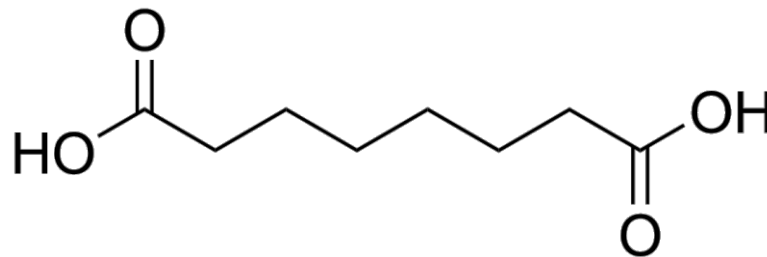
- Carnitine necessary for carnitine shuttle
 - Links with fatty acids forming acylcarnitine
 - Moves fatty acids into mitochondria for metabolism
- Muscle weakness, cardiomyopathy
- **Low carnitine and acylcarnitine levels**



MCAD Deficiency

Medium Chain Acyl-CoA Dehydrogenase

- Poor oxidation 6-10 carbon fatty acids
- **Dicarboxylic acids** 6-10 carbons in urine
 - Seen when beta oxidation impaired
- **High acylcarnitine levels**



Dicarboxylic Acid

Hypoketotic Hypoglycemia

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graph TD; A[Hypoketotic Hypoglycemia] --> B[Carnitine Deficiency]; A --> C[MCAD Deficiency];
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Carnitine Deficiency

Low carnitine levels
Low acyl-carnitine levels

MCAD Deficiency

High acylcarnitine levels
Dicarboxylic acids

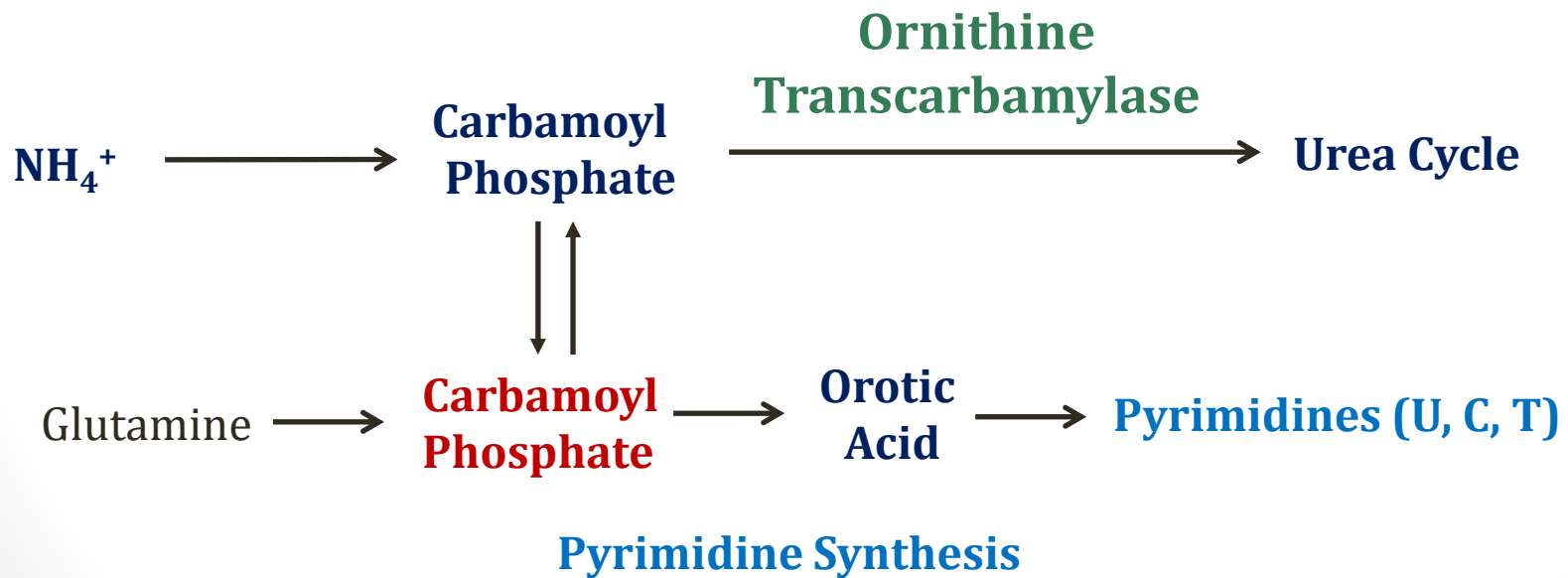
Urea Cycle Disorders

- Onset in newborn period (first 24 to 48 hours)
- Feeding → protein load → symptoms
- Poor feeding, vomiting, lethargy
- May lead to seizures
- Lab tests: **Isolated severe hyperammonemia**
 - Normal < 50 mcg/dl
 - Urea disorder may be > 1000
- No other major metabolic derangements

OTC Deficiency

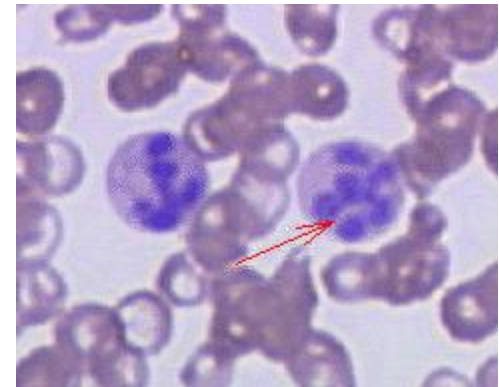
Ornithine transcarbamylase deficiency

- Most common urea cycle disorder
- ↑ carbamoyl phosphate
- ↑ orotic acid (derived from carbamoyl phosphate)



Orotic Aciduria

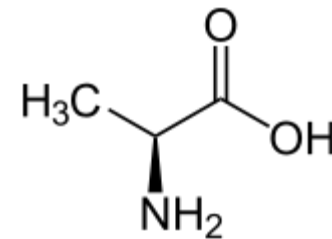
- Disorder of pyrimidine synthesis
- Also has orotic aciduria
- **Normal ammonia levels**
- No somnolence, seizures
- Major features: Megaloblastic anemia, poor growth



Megaloblastic Anemia

Mitochondrial Disorders

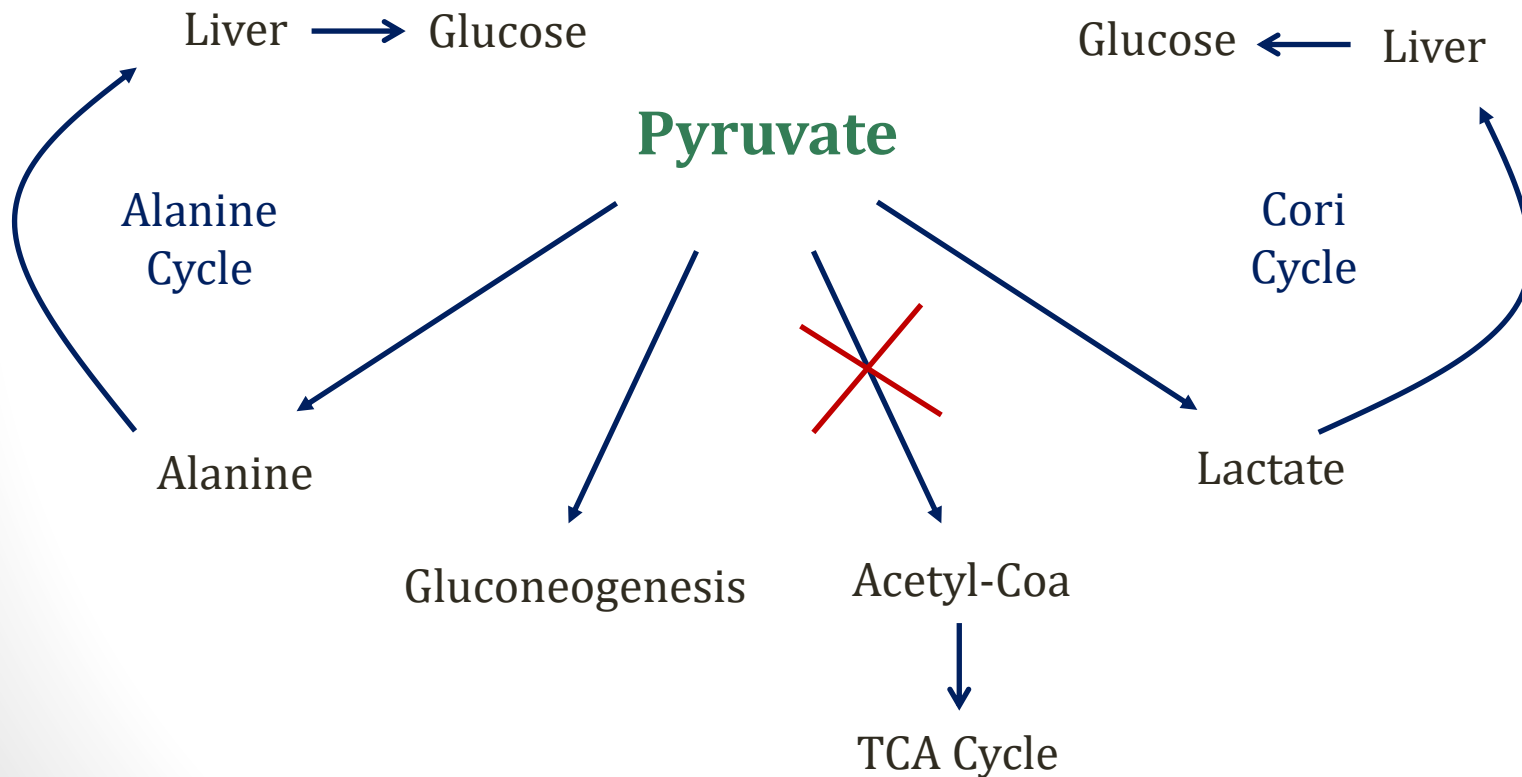
- Inborn errors of metabolism
- Loss of ability to metabolize pyruvate → acetyl CoA
- All cause **severe lactic acidosis**
- All cause **elevated alanine** (amino acid)
 - Pyruvate shunted to alanine and lactate
- Pyruvate dehydrogenase complex deficiency



Alanine

Pyruvate

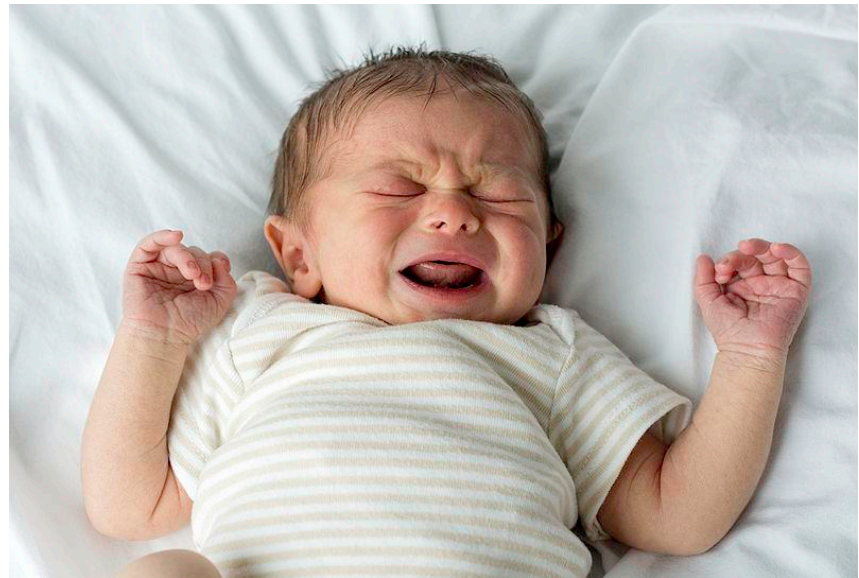
- End product of glycolysis



PDH Complex Deficiency

Pyruvate Dehydrogenase

- Pyruvate shunted to alanine, lactate
- Key findings (infancy):
 - Poor feeding
 - Growth failure
 - Developmental delays
- Labs:
 - Elevated alanine
 - Lactic acidosis
 - No hypoglycemia



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